Introduction to Agents

Guided reading notes — Google Cloud Whitepaper, November 2025

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> **How to use this document:** Read each section of the whitepaper, then fill in the notes on the matching page here before moving on. Your answers save automatically to this browser. At the end, answer the comprehension questions from memory — treat them like a short quiz.

pp. 6–7 From Predictive AI to Autonomous Agents

Read these pages, then complete the notes below before continuing.

The old AI paradigm vs. the new one — what is the core difference?

Old: passive, discrete, requires human direction at every step. New: ___

New: Software that is able to solve problems, execute complex tasks by making plans + taking actions.

Complete this sentence in your own words:

An agent is not simply a model in a static workflow — it is a complete application, making plans & taking actions to achi

What does the whitepaper mean by the shift from "prompt engineering" to "context engineering"?

n/a -> prompt engineering = static. context engineeering = dynamic

One question or observation from this section you want to come back to:

n/a

pp. 8–10 Introduction to AI Agents — Core Architecture

Fill in each component card as you read. Use the exact metaphors the paper uses, then add your own interpretation.

MODEL — "THE BRAIN"

What is its role?

Core LM that acts as agent's central reasoning engine to process info; dictates agent capabilities.

What does "agentic system as context window curator" mean here?

The language model can only see what is in its current context window. The agent, then, packages needed info for each step of process so the model sees only most relevant info.

TOOLS — "THE HANDS"

What is their role?

Connect agent reasoning to outside world to enable actions beyond text generation.

Two types of things tools can do:

1. Retrieval
2. Action

ORCHESTRATION — "NERVOUS SYSTEM"

What does it govern?

The agent's operational loop (planning, memory, reasoning strategy execution)

Examples of reasoning techniques it uses:

- Chain of thought: A set of intermediate reasoning steps to improve LLM complex reasoning
- ReAct

DEPLOYMENT — "BODY & LEGS"

Why does this matter beyond a laptop prototype?

Makes it reliable + accessible

How can other agents access a deployed agent?

Graphical interface or via an Agent-to-Agent API

The developer role is compared to a "director" rather than a "bricklayer." What changes in how you think about building software?

The dev sets the scene, selects the cast and provides the necessary context to guide the autonomous actor to deliver the intended performance.

pp. 10–13 **The Agentic Problem-Solving Process (5 Steps)**

Label and summarise each step as you read. Then apply the loop to the order-tracking example in the paper.

1	2	3	4	5
GET THE MISSION	SCAN THE SCENE	THINK IT THROUGH	TAKE ACTION	OBSERVE & ITERATE

Specific high level goal provided by a user	Check env to gather context = orchestration layer accessing relevant resources	Agent analyzes MISSION against SCENE to devise a plan that consists of a chain of reasoning	Selects + invoke appropriate tool - agent acting on the world	Agent observes action outcome, adds to memory/context, repeats from step 3 for next step

Walk through the order #12345 example step by step in your own words:

1) user wants order status
2) agent checks env to access relevant resources: order in internal database, external carrier api, etc.
3) agent formulates plan to (1) identify order (2) track order and (3) synthesize info for user
4) agent executes step (1) by calling find_order("12345") tool and getting order record + tracking number
5) agent saves this information in memory and repeats 4-5 for steps (2) and (3)
FINAL) after agent synthesizes info + generates response

pp. 14–18

A Taxonomy of Agentic Systems (Levels 0–4)

Fill in the table as you read each level. The last column is yours — invent an example not in the paper.

Level	Name	Key capability added	Your own example
L0	Core Reasoning	LM operates in isolation, working off pre-trained knowledge w/out tools & memory, etc.	Can answer history of US politics, but not result of election last night if out of training set
L1	Connected Problem-Solver	Agent has access to xternal tools; uses 5-step loop given previously	Can now answer above question
L2	Strategic Problem-Solver	executes multi-part goals through context engineering, accuracy + performance depends on model's ability to select, package + manage most relevant info for each step of the plan	"Find a good coffee shop halfway between my office and my client's office

Level	Name	Key capability added	Your own example
L3	Collaborative Multi-Agent	move away from one agent to multi-agent specialist team, strength in division of labor. agents treat other agents as tools	As "Project Manager" agent receiving a "Mission": "Launch our new 'Solaris' headphones." Delegates to MarketResearchAgent, MarketingAgent, WebDevAgent
L4	Self-Evolving System	Can expand multi-agent team independently and evaluate gaps in capabilities and dynamically improve	"Project Manager" agent, tasked with the 'Solaris' launch, might realize it needs to monitor social media sentiment, but no such tool or agent exists on its team.

What is "context engineering" and why does it become the defining skill at Level 2?

Context engineering is about packaging the most relevant information at every step to get the best results. In Level 2, we are able to execute multi-part goals through context engineering

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In the coffee-shop example (p.16), what exactly is the context engineering move the agent makes?

Synthesizes output from step 1 (finding the halfway point between the two offices as Milibrae, CA) to find the best coffee shops in Milibrae. In step 2 the agent takes the output of step 1 (Millbrae, CA) and automatically constructs a new, focused query rather than repassing all OG reasoning.

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pp. 19–26 Core Architecture in Practice — Model, Tools, Orchestration

These pages go deeper into how you actually build each layer. Note the concrete design advice.

Why is choosing the model with the best benchmark score a mistake for agents?

bc success not determined by these benchmark scores, but rather by picking a model that excels at agentic fundamentals: reasoning, tool use, etc. The "best" model is the one that sits at the optimal intersection of quality, speed, and price for your specific task.

What is model routing, and what problem does it solve?

The paper gives the example of Gemini 2.5 Pro for planning vs. Gemini 2.5 Flash for simpler tasks.

A way to optimize performance and cost by giving complex tasks to better models and simple tasks to cost-effective, simple models. Also helps in handling diverse data types! In addition, prepares your agent for changing model/AI landscape

The two categories of tools and how they differ:

Retrieval tools (grounding)

Access up-to-date info. RAG gives agent a "library card" to query external knowledge oft stored in Vector Databases or Knowledge Graphs. Structured data supported by Natural Language to SQL (NLSQL) tools to query databases & answer analytic questions. GOOD WAY TO REDUCE HALLUCINATIONS!

Action tools (executing)

- api calls
- send emails
- schedule meetings
- update CRM
- write + exec code
- use human in the loop (HITL) tools to pause workflow + ask for confirmation, ensuring a person in involved in the critical decisions

What is MCP (Model Context Protocol) and why is it becoming popular alongside OpenAPI?

MCP are open standards, offer simpler discovery + connection to tools.

A production-grade orchestration framework must be three things. Define each:

Open means: Allows you to plug in any model/tool to prevent vendor lock in

Precise means: Non-deterministic reasoning of an LM is governed by hard-coded business rules.

Observable means: Generate detailed traces + lods to expose model reasoning trajectory

Three multi-agent design patterns and when to use each:

Coordinator: DYNAMIC TASKS: A manager agent that analyzes a complex request, segments the primary

Sequential: LINEAR WORKFLOW: Digital assembly lines

Iterative Refinement: QUALITY + SAFETY: Generator agent & critic agent feedback loop

pp. 27–31

Agent Ops — Evaluating, Debugging & Improving

This section introduces a new operational discipline. Capture the key practices as you read.

Why can't you use a simple pass/fail unit test for agents?

Agent's response is probabilistic by design.

"LM as Judge" — what is it and what does a good rubric check for?

Use a model to assess the agent's output against a predefined rubric. The rubric can check for things like Did it give the right answer? Was the response factually grounded? Did it follow instructions? during sample scenarios.

What is a golden dataset and how should you build one?

A golden dataset is filled with ideal questions + correct responses, built from sample scenarios from existing production or development interactions with the agent. Covers the full breadth of use cases that users can. engage w/ + unexpected scenarios

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What does an OpenTelemetry trace show you that metrics dashboards cannot?

An OpenTelemetry trace is a high-fidelity, step-by-step recording of the agent's entire execution path (trajectory), allowing you to debug the agent's steps.

The Agent Ops feedback loop — how does a user bug become a permanent fix?

Think: collect → replicate → convert → prevent

By capturing this feedback, replicating the issue, and converting that specific scenario into a new, permanent test case in your evaluation dataset. This ensures you not only fix the bug but also vaccinate the system against that entire class of error ever happening again.

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pp. 31–34 Agent Interoperability — Humans, Agents & Money

MCP vs. A2A — what is the distinction between them?

MCP	A2A
MCP focuses on solving transactional requests.	Solves the problem of agent discovery and communication by acting as a "universal handshake" for the agentic economy.

What is an Agent Card and what problem does it solve?

Agent Card is a JSON file with agent capabilities, network endpoint + security credentials to interact with it.

What trust problem arises when agents transact with money, and how do AP2 and x402 address it?

Essentially, the question is of accountability - who is responsible for agent decision making/wrongs. AP2 is an open protocol acting as a definitive language for agentic commerce, extending protocols like A2A by including digital "mandates" with proof of user intent. x402 is an open internet payment protocol that enables micropayments.

pp. 34–41

Security, Identity & Governance

The trust trade-off — what are the two risks you balance when giving an agent power?

Risk 1: Rogue actions: unintended & harmful actions

Risk 2: Sensitive data disclosure

The defense-in-depth approach has two layers. Describe each:

Layer 1: Deterministic guardrails (ex: a policy engine that blocks any purchase over $100)

Layer 2: Reasoning-based defenses (This involves training the model to be more resilient to attacks and empl

Agents are called a "third category of principal." What does this mean for how you manage identity and permissions?

Humans use OAuth/SSO, services use IAM or service accounts, agents also need "digital passports." Ex: SPIFFE. Gives agents delegated authority to take actions on behalf of users.

What is "agent sprawl" and what two functions does a central governance control plane serve?

Agent sprawl is the complex network of interactions and potential vulnerabilities that come from many agents. A central governance control plane manages the chaos through runtime policy enforcement and centralized governance (as a source of truth).

pp. 42–50

How Agents Learn, Evolve & Advanced Examples

What is "agent aging" and why is it a problem at scale?

Without adapting in a dynamic environment where policies, tech, data formats are always changing, the agent's performance will degrade. Manually updating a fleet of agents to keep pace w/ these changes is slow.

Two categories of adaptation techniques and what each involves:

Enhanced context engineering: System continually refines prompts, few-shot examples, info from memory

Tool optimisation & creation: Can identify gaps in capabilities and fill them through updating/new tools

What makes the "Agent Gym" different from ordinary in-line learning?

Off production, simulation environment, advance synthetic data generators, adopt new tools, etc.

Co-Scientist and AlphaEvolve — one key thing each illustrates about advanced agents:

Co-Scientist shows: Spawns an agent ecosystem collaborating with each other

AlphaEvolve shows: Generator + evaluator feedback loop creates continuous improvements.

Comprehension Questions

Close the whitepaper and answer from memory. These test understanding, not recall.

QUESTION 1

Explain the "context window curation" metaphor in plain English. Why does the whitepaper call this the essence of what an agent is?

Essentially, context window curation means controlling what information + tools the agent has access to, so as to get the best output/action. This is the essence of what an agent is as agents take actions based on inputs. However, if another agent can determine the best info to give, we can create even more complicated systems.

QUESTION 2

A product manager says "let's just pick the model with the best leaderboard score." What is the flaw in this reasoning, and what should they do instead?

The leaderboard score does not correspond with the agent that is best suited to the specific task at hand; at reasoning, etc. Instead, they should pick the agent that fits their cost needs and is most suited to the specific task.

QUESTION 3

What is the difference between MCP and A2A? Give a concrete scenario where each would be the right choice.

MCP connects an agent to tools & data sources, and A2A connects agents to other agents for collaborative problem solving. Finding a good coffee shop in a certain area is a MCP task because you would need tools like Google Maps to do such a thing. A2A would be used for bigger tasks, like the Research Assistant agent described in the whitepaper.

QUESTION 4

Describe the "defense-in-depth" approach to agent security. Why can't you rely on the model's own judgment alone?

There are many security concerns with relying on the model's own judgement; data poisoning, prompt injection, etc. can shape what the model is doing in negative ways. Defense-in-depth is multi-step, and prevents agents from making harmful/incorrect decisions and taking certain kinds of actions by limiting credentials, checking for prompt injects, etc. Layer 1 = deterministic hardcoded guardrails (policy engines, hard rules outside the model), Layer 2 = AI-powered reasoning defenses (guard models, adversarial training).

QUESTION 5

You are evaluating an agent and notice quality has dropped over time. Walk through the Agent Ops process for diagnosing and fixing the problem — from detecting the issue to preventing it from recurring.

Use the specific tools and processes described in the whitepaper.

Use OpenTelemetry traces to see things like what prompt was sent, what tool was chosen, what parameters were generated, and what came back. The Agent Ops process involves capturing specific this feedback on this degradation, replicating the issue, and then shaping the agent build itself to ensure this does not happen again, both through prompt refinement and through capturing specific scenarios, failures and providing the agent with scenario in training datasets so the system is vaccinated against the class of issues.

QUESTION 6

Your company wants to build an agent that can autonomously research competitors, draft a report, and email it to stakeholders weekly. What agent level is this, which multi-agent design pattern fits, and what are the top two security concerns you must address?

This is a Level 3 agent, and the "manager agent" pattern fits best. The manager agent can develop a plan that is delegated to a Research Agent, a Report Creation Agent, and an Email Agent to execute this plan. One security concern is prompt injection that can get proprietary company info from the agent. Another is data poisoning that leads to false information being provided to stakeholders.

QUESTION 7

In your own words, why does the whitepaper argue that success in agent development depends more on "engineering rigor applied to the whole system" than on the initial prompt?

The initial prompt is just that - initial. Engineering rigor comes into the refinement of the agent, the security developed, testing, etc. An LM's flexibility (its ability to do anything) is also the source of its unreliability, so success comes from robust tool contracts, error handling, context management, and evaluation, not from crafting a better prompt.

Reflection — after finishing

One sentence: the most important idea in this whitepaper.

n/a

What most surprised or challenged an assumption you had going in?

n/a

Three things you want to explore further:

1. n/a
2. n/a
3. n/a

One concrete next action in the next 2 weeks:

n/a

Guided notes for "Introduction to Agents and Agent Architectures" — Google Cloud, November 2025